# Louis Antoine

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**WWW:** https://alovladi007.github.io/louis-antoine-portfolio/index.html

## Summary

Experienced in front end semiconductor fabrication, with hands on expertise in photolithography, thin film deposition (PVD/CVD/ALD), dry etch, and metrology systems. Skilled in tool qualification, recipe optimization, chamber matching, and preventative maintenance across leading platforms (ASML, AMAT). Proficient in SPC, FDC, DOE, and root cause analysis to drive process stability, yield improvement, and equipment uptime. Strong background supporting high volume manufacturing (HVM) and NPI environments.

## Skills

- Thin film deposition process development (PVD, CVD)
- Image processing and analysis (MATLAB, Python, OpenCV, scikit-image)
- Photolithography process development and optimization
- EUV / DUV lithography tool operation and calibration
- Tool qualification and characterization (install, baseline, spec check, system readiness)
- Optical System Design and Alignment.
- Image acquisition and sensor operation
- Optical Proximity Correction (OPC) and Resolution Enhancement Techniques (RET).
- Reticle and pellicle management (binary masks, phase shift masks (PSM))
- Metrology and data analysis (line edge roughness (LER), swing curves, focus-exposure data)
- Yield enhancement (DOE, FMEA, 5 Whys, and Pareto analysis)
- Facilities and subsystem knowledge (familiarity with vacuum systems, gas delivery MFCs, cooling loops, chemical/vapor lines, and electrical diagnostics, oscilloscopes, multimeters)
- Subsystems: vacuum, RF/plasma, motion control, thermal systems
- Process monitoring and metrology (ellipsometry, XRR/XRD, profilometry, sheet resistance)
- Defect inspection and metrology (CD-SEM, overlay metrology, AFM, and scatterometry)
- Wet etch chemistry and process development documentation and compliance
- Cleanroom wet bench operation and safety
- Documentation and compliance

## Experience

**GRADUATE RESEARCH ASSISTANT / TA** | 10/2022 - 04/2024

**University of Connecticut - Storrs, CT**

- Applied ML methods to data analysis and experimental optimization.
- Modeling and Evaluation of Gallium Nitride (GaN)-Based Power Devices for High-Frequency Switching Applications
- Silicon Carbide (SiC): Developed detailed models of SiC MOSFETs to evaluate their performance in harsh, high-voltage environments.
- Monte Carlo Simulation vs. Double Gaussian Model: Accurate modeling of electron scattering and energy deposition in EUV masks, outperforming analytical proximity correction models in complex 3D absorber structures.
- Supported semiconductor and optoelectronics labs, assisted with precision measurements, front-end, and back-end (interconnect, dual Damascene) processing technologies.

- lithography (Optical Pattern Formation, Photoresists, Wafer Steppers, and Scanners (see ASML Work Experience), Overlay, Overlay, Immersion Lithography and Limits, Extreme Ultraviolet Lithography).
- Etching technologies and chemical-mechanical polishing (Wet Etching, Dry or Plasma Etching, Atomic Layer Etching). Etchback process using photoresist and spin on glass (SOP), familiarity with Applied Materials Reflection CMP machine, A.
- Deposition technologies (CVD, ELO, LPCVD, MOCVD, PECVD, ALD, and molecular beam epitaxy (MBE)).
- Provided guidance to undergraduate students working on independent projects related to the main research focus.

**SR TECHNICIAN / PRODUCTION TESTER** | 05/2021 - 09/2022

**Asml Us - Wilton, CT**

- Used standardized tools and electronic equipment (torque wrenches, electric and pneumatic screwdrivers, ball drivers, lifting equipment, computers, oscilloscopes, voltage and current probes, multimeters, micrometers, and calipers) to assemble, install, remove, or perform testing and troubleshooting on an optical metrology system. Used optical fibers, spectrometers, and power meters to obtain and analyze results.
- Experienced with electro-optical and/or optical-mechanical systems, as well as imaging and testing. Familiarity with the YieldStar 375 F, 380 G, and 1385 Optical Metrology System for process monitoring and control, as well as for lithography system stability and matching. Performed sub system assembly, optical alignment, functional testing, equipment calibration, station setup, and process development support.
- Sensor alignment, UIA alignment, focus branch, Z-stage installation, pre qualifications, Pupil Branch, Alignment Branch, Reference Branch, Cables, and Covers, final qualifications.
- Final Qualifications: Measured sensor performance; calibrations, alignment of branch optics, objective lens performance, and diffraction based overlay measurements.
- Assisted with making recommended changes in design, methods, and procedures to enhance product quality.
- Developed spreadsheets using Excel to track key performance indicators.
- Worked in a clean room environment, and adhered to strict safety procedures, utilizing PPE, and embracing ASML 5S+1 requirements. Wore coveralls, hoods, booties, safety glasses, and gloves for the entire duration of 12 hour night shifts.
- Performed root cause analysis on production issues, developed corrective actions with the SAP system, Opti Angle, YieldStar software, and monitored results.
- Participated in cross functional teams with manufacturing, quality, and procurement departments.

**UNDERGRADUATE STUDENT RESEARCH ASSISTANT** | 05/2020 - 08/2020

**University of Connecticut - Storrs, CT**

- Utilized EIT for slow light and quantum memory demonstrations in vapor cells and cold atom ensembles.
- Modeled Λ type three level systems using density matrix formalism to simulate optical susceptibility under varying laser detuning and intensity.
- Developed laser locking and modulation systems for EIT experiments using acousto-optic modulators (AOMs), and electro-optic modulators (EOMs).

**PATIENT ADMINISTRATION SPECIALIST** | 01/2016 - 11/2017

**United States Army - Fort Stewart, GA**

- Utilized the computerized Resource and Patient Management System (RPMS) and Electronic Health Record (EHR) system to update patient records, transmit prescriptions, and transfer files.
- Maintained strict patient data procedures to comply with HIPAA laws, and prevent information breaches.

## Education and Training

**University of Connecticut - Storrs, CT | Master of Science**

Electrical Engineering, 12/2024

**University of Connecticut - Storrs, CT | Bachelor of Science**

Physics, 08/2020

**CT State Community College Housatonic - Bridgeport | Associate of Science**
Engineering Science, 12/2015

## ADDITIONAL EXPERIENCE

- **Professional Driver (Uber/Lyft)**
  Storrs, CT
  Jan 2018 – Apr 2021
(While completing my undergraduate degree)
- **Cashier Specialist**
  Walmart, Norwalk, CT
  Oct 2011 – Nov 2015

## Certifications

- Six Sigma Green Belt
- Google IT Automation with Python
- Google Advanced Data Analytics Professional Certificate
- MATLAB Programming for Engineers and Scientists Specialization